

Education, Skills, Aspirations, And Occupational Sorting

Gender Study — Week 3

Teaching deck draft

Central question

- How do education, skills, aspirations, and early occupational choices generate later gendered labor-market outcomes?
- This is labor economics because sorting shapes skill prices, job access, career ladders, wage growth, authority, and job quality.
- Week 3 discipline: separate early choices from later adjustment frictions.

Week 2 to Week 3 bridge

- Week 2: household allocation, care, fertility, and expected family roles.
- Week 3: those expectations can affect schooling, major choice, aspirations, internships, search, and first jobs.
- A flexible first track can be an early choice shaped by expected later constraints.

Why sorting is not one mechanism

Channel	Object
Early choices	schooling, major, track, skills, aspirations, role models, information
Entry sorting	first occupation, sector, employer, task, ladder, schedule
Later adjustment	switching costs, promotion gates, flexibility constraints, persistence
Interactions	anticipated future constraints shape early choices; early choices raise later costs

Conceptual decomposition

$$G_Y = \Delta_Y^{early} + \Delta_Y^{entry} + \Delta_Y^{adjust} + \Delta_Y^{returns} + \Delta_Y^{interactions}$$

- Y : earnings, wages, authority, hours, promotion, job quality, or welfare.
- Early: skills, schooling, aspirations, beliefs, information.
- Adjustment: persistence, switching, family timing, mobility, promotion.
- Decomposition is not identification; it is a map of claims.

$$o_i^0 = \arg \max_{o \in \mathcal{O}} \left\{ \mathbb{E}_{i0} \left[\sum_{t=0}^T \beta^t w_{ot}(H_{it}) \right] + A_{io} - C_{io} - D_{io}^e - \kappa_{io}^e \right\}$$

- A_{io} : aspirations, identity fit, perceived belonging.
- D_{io}^e : expected discrimination or exclusion.
- κ_{io}^e : anticipated future adjustment costs.

Early choices channel

- Field of study, training, courses, internships, and first occupation.
- Role models and mentors shift information and perceived belonging.
- Teacher signals and stereotypes shape perceived comparative advantage.
- Competition, risk, and aspirations affect entry into high-variance tracks.
- Anticipated discrimination changes choices before employer treatment is observed.

Aspirations, competition, information, role models

Mechanism	Observed margin
Female role-model exposure	economics major choice
Competitiveness measure	later track or career choice
Teacher feedback / bias	course taking, achievement, field choice
Information about returns or fit	applications, major choice, intended field

- Name the identifying variation before interpreting the result.

Anticipated discrimination and field choice

- Expected treatment can enter the supply-side choice problem.
- A student may avoid a field because she expects worse hiring, mentoring, grading, promotion, or workplace climate.
- Observed sorting is then not a pure preference measure.
- Beliefs designs should name whether they measure expected treatment, shift beliefs, or observe intended choices.

$$V_{it}(o) = \max \left\{ u_{it}(o) + \beta V_{i,t+1}(o), \max_{o' \neq o} [u_{it}(o') - K_{i,t}(o, o') + \beta V_{i,t+1}(o')] \right\}$$

- $K_{i,t}$: retraining, lost seniority, networks, credentials, relocation, flexibility constraints.
- Later adjustment is a separate channel from initial field choice.

Later adjustment and inertia

- First placements can define training, networks, tasks, schedules, and promotion gates.
- Early-career shocks can alter long-run labor and family trajectories.
- High-skill career ladders often reward continuous hours and early promotion races.
- Direct barriers can make movement out of early tracks costly even when workers want to switch.

Early-career path dependence

Object	Evidence to look for
First job or specialty	quasi-random assignment, placement shock, cohort rules
Career ladder	wage growth, promotion, task assignment, retention
Flexibility constraint	hours, schedule, family timing, job switches
Persistent mismatch	later occupation, earnings, authority, job quality

Channels can interact

- Anticipated future constraints can change initial major or occupation choice.
- Early choices build specific skills, credentials, networks, and seniority.
- Family timing can raise the cost of retraining or switching later.
- Firm promotion systems can amplify small early differences into large career gaps.

Short bridge to Week 5 norms

- Norms shape aspirations, acceptable jobs, self-promotion, mobility, and perceived feasibility.
- Domestic norms can affect expected care responsibilities before career entry.
- Workplace norms can affect whether technical, competitive, or authority-bearing roles feel accessible.
- Week 3 flags the bridge; Week 5 studies norms and institutions directly.

Methods map

Design		Identifying variation	Observed margin
Role-model intervention		classroom exposure	major or field choice
Teacher-bias design		teacher signals / grading environment	course, track, field
Expectations survey		measured or shifted beliefs	intended or realized choice
Early-career experiment	quasi-	placement, shock, assignment	persistence, switching, wage growth
Long panel		lifecycle histories	timing, path dependence
Decomposition		accounting variation	shares across occupations or fields

- Reproduce: synthetic role-model intervention inspired by Porter and Serra.
- Diagnose: treatment, outcome, identifying variation, margin, and limitation.
- Transfer: synthetic competitiveness and track-choice exercise inspired by Buser, Niederle, and Oosterbeek.
- Optional frontier prompt: anticipated discrimination or early-career setbacks.

- Which early-choice interventions produce durable labor-market outcomes?
- How do role models change information, belonging, expected returns, and discrimination beliefs?
- Which early-career shocks reveal true adjustment frictions?
- What data connect education, first job, firm assignment, promotion, and family timing?

- Week 3 explains who enters which fields and early career ladders.
- Week 4 moves inside firms: hiring, evaluation, pay-setting, promotion, authority, and retention.
- The next question is how firms transform worker sorting into pay, authority, and career outcomes.

- Gendered sorting has early-choice and later-adjustment channels.
- Observed sorting is not a pure preference or pure constraint statistic.
- Credible evidence names the variation, the observed margin, and the mechanism.
- Week 3 links household expectations to firm-side career dynamics and the later norms lecture.